Local Topological Fragility in Game of Life and HighLife

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Life-like cellular automata exhibit *local topological fragility:* the initial density of weakly-connected live cells predicts future loss of fine-scale connected components. Across Conway's Game of Life (GoL) and HighLife, two grid sizes ($L \in \{64, 128\}$), four densities, and seven forecast horizons, a sixteen-statistic topology battery identifies a predictive hierarchy: initial component count $C(0)$ (mean residual $R^2 = 0.352$), small-component counts (0.330), singleton/orthogonal-degree-zero cells (0.274), and *embedded isolates*—alive cells with no orthogonal and at least one diagonal live neighbour—as the mechanistically transparent motif ($R^2 = 0.234$; prestate $R^2 = 0.241$; min 0.175 at $k = 200$; shuffle ≈ 0.0008).

The effect is target-specific: both shuffle increments are near zero ($\Delta R^2 < 0.001$), and the residual slope $\hat{\beta}_{\text{iso}}(k)$ is negative in all 112 condition-horizon combinations ($k \in \{1, 5, 10, 25, 50, 100, 200\}$; CV $\lesssim 0.17$). Mechanistically, any orthogonal-degree-zero cell with Moore degree 0 or 1 dies at $t+1$ as an exact S_{23} rule consequence; multi-diagonal isolates more often survive-connected. Both pathways reduce the component count; a local-window feature captures this ($R^2 = 0.538$). The standardised mechanism transfers across held-out conditions ($R_z^2 = 0.544$), and raw amplitude is calibrated from (L, ρ) (LOO $R^2 = 0.970$).

Keywords: cellular automata, coarse-graining, embedded isolated cells, temporal response, mechanism transfer

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I. INTRODUCTION

Conway's Game of Life (GoL)[1] and HighLife are among the most studied cellular automata, valued for their deterministic dynamics, rich emergent structure, and computational tractability. Earlier work established edge-of-chaos properties[2,3] and statistical complexity[4] of Life-like rules, and identified the role of coherent structures[5] and computational irreducibility[6] in deterministic CAs. Local causal-state methods[4,5] identify predictive-equivalence partitions from past/future lightcones; our work instead asks which simple hand-defined structural classes are privileged for a specific coarse-grained target observable. The partial predictability of a chosen observable demonstrated here is consistent with computational irreducibility[6]: we do not claim to predict the full future CA configuration, only one low-dimensional target statistic.

We identify *local topological fragility*—the initial density of structurally isolated or weakly-connected live cells—as a target-indexed property of Life-like CA states that predicts future fine-component loss. A battery of sixteen topology statistics reveals a predictive hierarchy: initial component count $C(0)$ is the strongest aggregate predictor; small-component and singleton-component (orthogonal-degree-zero) counts are the strongest local-structure predictors; and the count of *embedded isolated cells* (iso_count)—alive cells with no 4-connected live neighbour and at least one diagonally adjacent live neighbour—is the most analytically transparent motif within that family. Embedded isolates sit at the topological boundary of 4-connectivity: orthogonally isolated, diagonally tethered, they are the mechanistically cleanest fragile connectivity nodes in the grid, because their one-step fate under the S_{23} rule has a simple analytic form. Earlier observer-scale diagnostics from the same research program motivated this target-specific analysis; they are documented in the public repository.

The main results, established across GoL and HighLife, two grid sizes ($L \in \{64, 128\}$), four density bands ($\rho \in \{0.20, \ldots, 0.35\}$), and seven prediction horizons, are: (1) a topology-baseline battery of 16 statistics shows that local fine-topological fragility—led by $C(0)$, small-component counts, and singleton/orthogonal-degree-zero counts—strongly predicts fine-net change, with embedded isolates as the most analytically transparent motif; (2) iso_count predicts future fine-component loss with mean prestate $R^2 = 0.241$, sustained above 0.175 at $k = 200$; (3) the effect is target-specific—incremental lift for density, entropy, and coarse-block targets is near zero, as are both shuffle nulls; (4) the residual slope $\hat{\beta}_{\text{iso}}(k)$ is negative

in all 112 condition-horizon combinations, with CV $\lesssim 0.17$; (5) the mechanism is local component-context loss, not simple cell death, and transfers across all leave-one-out axes.

Figure 1 provides the entry point: the structural definition of an embedded isolate and the target-specific ΔR^2 pattern that singles out the fine-component target.

II. DEFINITIONS AND PROTOCOL

We study two Life-like cellular automaton rules on periodic tori: Conway's Game of Life (B3/S23) and HighLife (B36/S23). Grid sizes $L \in \{64, 128\}$ with toroidal boundary conditions. The *fine observer* counts 4-connected live-cell components (BFS on the wrapped grid); the *coarse observer* counts occupied 8×8 blocks. An *embedded isolate* is an alive cell for which all four orthogonally adjacent cells are dead and at least one of the four diagonally adjacent cells is alive (Fig. 1a). We denote the 4-connected component count at time t as $C(t)$ and define the fine-net change at horizon k as $\Delta C_k = C(t+k) - C(t)$.

The experiment uses **16** conditions:

$$\{\text{GoL}, \text{HighLife}\} \times \{64, 128\} \times \{0.20, 0.25, 0.30, 0.35\},$$

giving 2 rules$\times$2 sizes$\times$4 density levels = 16 independent condition–horizon batteries. Each condition uses 1000 independently drawn random initial states at the specified density ρ. We test $k \in \{1, 5, 10, 25, 50, 100, 200\}$, yielding 7 horizons $\times$ 16 conditions = 112 condition-horizon slope estimates.

The residual slope $\hat{\beta}_{\text{iso}}$ is estimated by ordinary least squares (OLS) with $N_{\text{live}}(0)$ (total live-cell count, proportional to ρ) as a nuisance covariate, so that iso_count is evaluated for its density-orthogonal contribution. This and all horizon-response regressions are fitted *condition-wise* (within each of the 16 batteries independently); the global $\bar{\beta}_{\text{iso}}$ reported in the text is the mean across conditions. Shuffle-null analyses (iso-count label permutation or target permutation) are also run within condition. Mechanism R^2 values are 5-fold cross-validated within condition. Slope confidence intervals use world-resampling bootstrap (≥ 300 draws per condition) rather than parametric OLS standard errors. Each initial grid is drawn independently, making worlds the natural bootstrap unit.

We use four distinct R^2 quantities throughout and distinguish them as follows. (a) *Ordinary R^2*: variance explained by an OLS model with no cross-validation; used for prestate fits.

(b) *Residual* R^2: R^2 of iso_count after the density covariate has been projected out; this is the within-condition prestate R^2 reported in Section III and Table I. (c) *Cross-validated* R^2 (5-fold, within condition): used for mechanism features in Section V. (d) ΔR^2: the difference $R^2(\text{density + predictor}) - R^2(\text{density only})$; used in the target-selection screen (Section III) to measure incremental lift over the density baseline.

All results are scoped to these two Life-like rules on periodic tori with $L \in \{64, 128\}$ and $\rho \in \{0.20, 0.25, 0.30, 0.35\}$. Low-density dynamics ($\rho \lesssim 0.10$), other Life-like rules, and non-random initial conditions are outside the present scope. We do not rerun experiments; all statistics are drawn from the fixed set of CSVs in `outputs/`.

III. TARGET-SPECIFIC SELECTION AND NON-LEAKY PRESTATE

A. Target-specific ΔR^2

Figure 1b shows the selection-principle battery: for each candidate predictor we measure $\Delta R^2 = R^2(\text{density + predictor}) - R^2(\text{density only})$ across six displayed targets, with iso_count, coarse block-variance, block entropy, and two shuffle nulls as predictors (see Appendix A for the residualisation protocol).

iso_count on the fine-net target yields $\Delta R^2 = 0.0012$, a positive incremental lift in both GoL and HighLife screens and in the global pooled screen. Both shuffle nulls are near zero ($\Delta R^2 < 0.001$), confirming the signal is not an artefact of the density-count correlation. iso_count adds essentially nothing for density, block-variance, block-entropy, or coarse-component targets: the uplift is concentrated on the fine-net target only. This specificity rules out explanations based purely on total alive-cell count or generic spatial statistics.

Two distinct R^2 *quantities.* $\Delta R^2 = 0.0012$ and the prestate $R^2 \approx 0.241$ reported in the next subsection answer different questions and should not be compared directly. The ΔR^2 screen asks whether iso_count adds *selective* information beyond the live-cell-count baseline *across multiple targets simultaneously*; because the baseline already absorbs most of the pooled variation, the incremental lift is numerically small by construction. The prestate R^2 evaluates the density-residualised fine-component target directly within each condition–horizon battery: how much of the remaining variation does $t = 0$ iso_count explain? That is the main effect-size estimate—iso_count accounts for roughly 0.241 of residual

fine-component-change variation on average across horizons, falling no lower than 0.175 at $k = 200$.

B. Non-leaky prestate recovery

A natural concern is whether iso_count is a proxy for information that requires knowing the one-step fate of individual cells. Figure 2 settles this: using *only* iso_count at $t = 0$—no fate features, no local-window information—the prestate model achieves mean $R^2 = 0.241$ across all horizons $k \in \{1, \ldots, 200\}$ (minimum $R^2 = 0.175$; shuffle ≈ 0.0008). Adding the full diagonal-class breakdown gives no substantial gain. The full multi-horizon iso-count signal is recoverable from the t=0 count alone.

IV. TEMPORAL RESPONSE CURVE

For each of the 112 condition-horizon combinations we estimate $\hat{\beta}_{\text{iso}}(k)$ and the corresponding residual R^2. We also compute an iso-shuffled null at every horizon.

All 112 slopes are negative; all 112 bootstrap slope confidence intervals are negative (= 1.00 at every horizon). The global mean slope moves modestly from -0.798 at $k = 1$ to -0.709 at $k = 200$ (range ≈ 0.09) while the CV stays below 0.17 throughout (Table I). Predictive power decays from $R^2 = 0.361$ at $k = 1$ to $R^2 = 0.175$ at $k = 200$, yet remains well above the null at every horizon.

Figure 3 shows the slope and R^2 curves. The slope plateau from $k = 5$ to $k = 200$ indicates that the mechanism acts primarily in the first step: once the initial fragile nodes are resolved at $t = 1$, the remaining dynamics do not strongly modulate the iso_count signal. The CV profile confirms robustness: condition-level spread is small relative to the global mean, and GoL and HighLife track each other closely despite different neighbourhood rules.

Figure 4 provides the complete picture: a 16×7 heatmap of all condition-horizon slopes, every cell shaded negative. There are no outlier conditions or horizons.

V. MECHANISM: LOCAL COMPONENT-CONTEXT LOSS

Having established *that* iso_count predicts fine-component change, we turn to *how* this occurs—what actually happens at $t \to t + 1$ when embedded isolates reduce the future

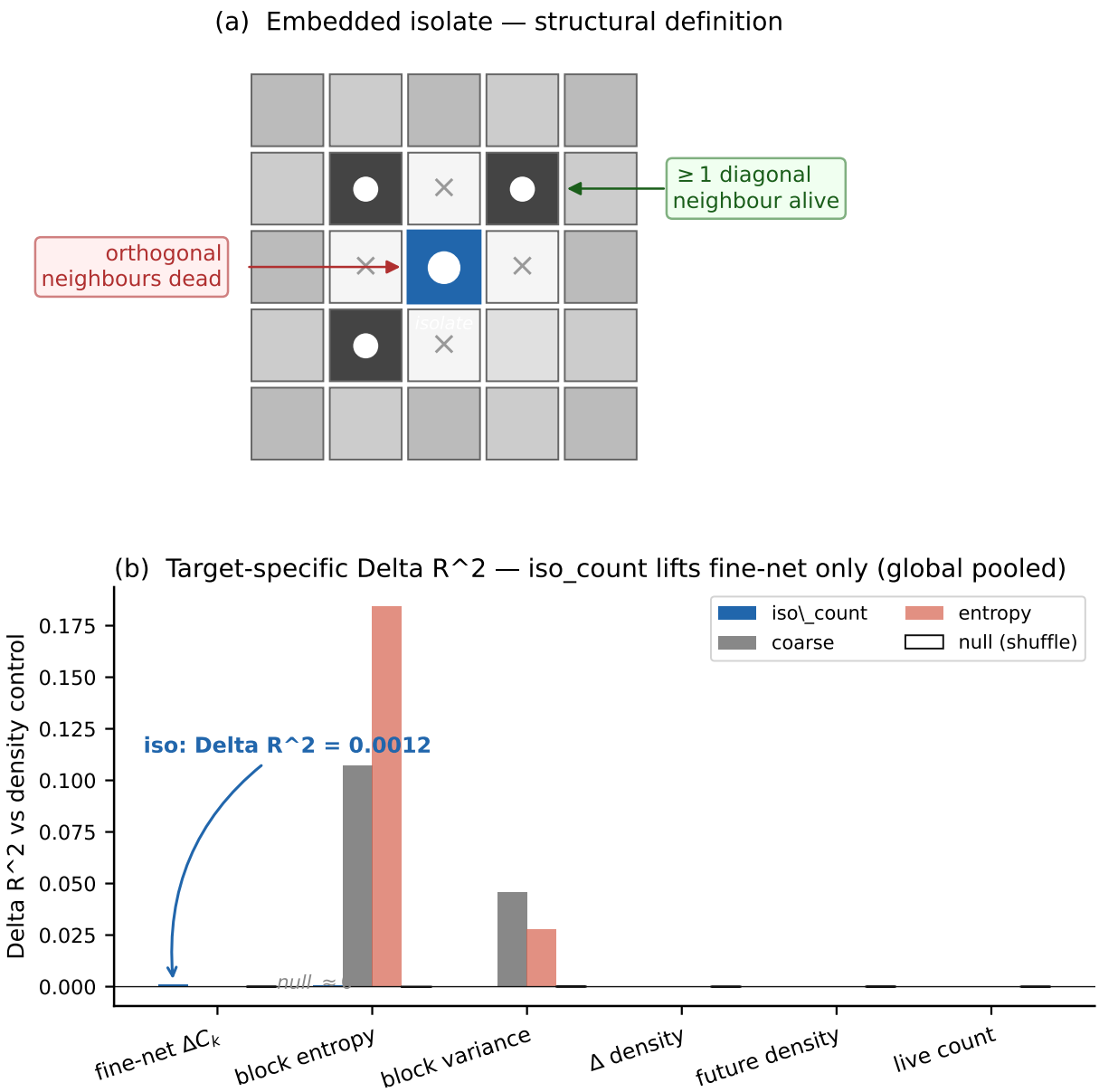

FIG. 1. **Structural definition and target-specific selection.** (a) An embedded isolate: an alive cell (filled circle) with all four orthogonal neighbours dead (×) and at least one diagonal neighbour alive (filled grey). This configuration places the cell at the most fragile locus of the local component topology. (b) ΔR^2 bars for iso_count and comparison predictors (coarse, entropy, null) on six displayed targets (global pooled; `target_delta_components` is numerically identical to fine-net and omitted). iso_count adds a positive incremental $\Delta R^2 = 0.0012$ for fine-net only; both nulls are near zero ($\Delta R^2 < 0.001$), confirming the signal is not a density artefact. The ΔR^2 bar is small because it measures incremental lift over a strong density baseline; the main residual effect size is the condition-wise prestate R^2 shown in Fig. 2.

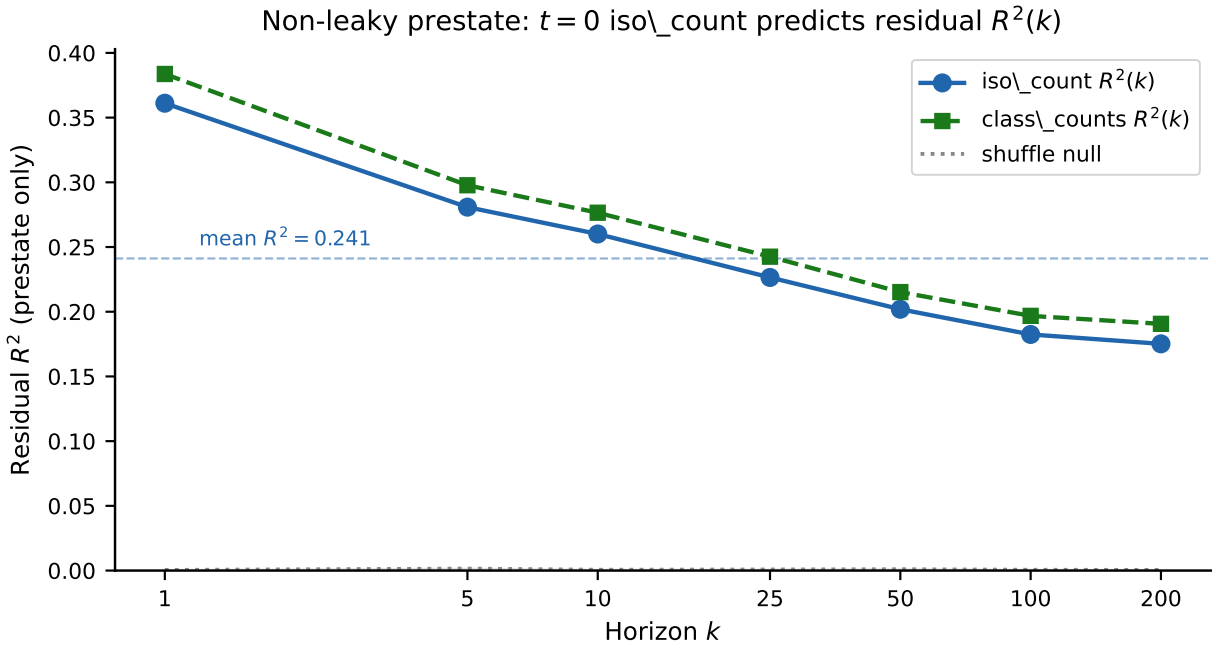

FIG. 2. **Non-leaky prestate recovery.** Using only $t = 0$ iso_count (no fate features, no local-window information), residual $R^2(k)$ stays above 0.175 at every horizon and averages 0.241 (dashed line). Adding class-count breakdown (green) gives marginal improvement. The grey band shows the shuffle null ($R^2 \approx 0.0008$). The tested iso-count signal is recoverable from the t=0 statistic alone.

TABLE I. Temporal response summary (16 conditions, 1000 worlds each; bootstrap CIs, ≥300 draws). All slopes and slope confidence intervals are negative at every horizon (frac_CI_neg = 1.00).

k	$\bar{\beta}_{\text{iso}}$	CV	$\bar{R}^2$	Null R^2	n_{cond}
1	−0.798	0.166	0.361	0.001	16
5	−0.710	0.146	0.281	0.001	16
10	−0.712	0.126	0.260	0.001	16
25	−0.703	0.143	0.226	0.002	16
50	−0.699	0.138	0.202	0.002	16
100	−0.701	0.137	0.182	0.001	16
200	−0.709	0.123	0.175	0.001	16

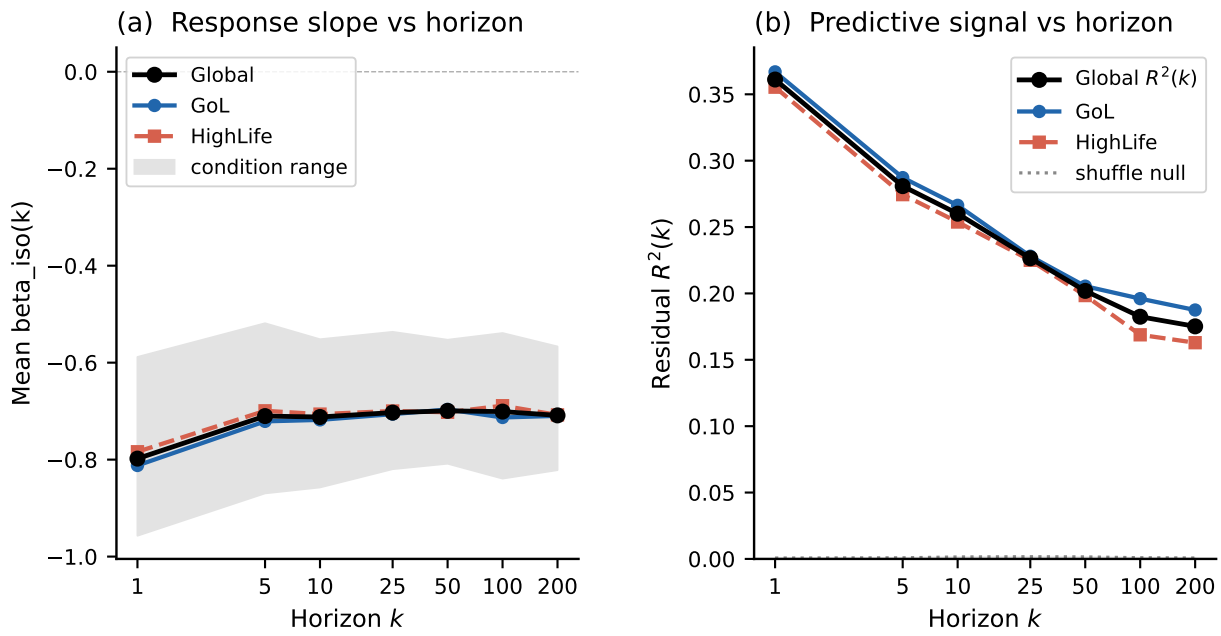

FIG. 3. **Temporal response: slope and predictive signal.** (a) Residual slope $\hat{\beta}_{\text{iso}}(k)$ versus horizon k for GoL (blue circles), HighLife (orange squares), and global mean (black); shaded region shows condition-level spread. The slope is negative and stable from $k = 1$ to $k = 200$. (b) Residual $R^2(k)$ versus k; grey band marks the shuffle null. Predictive power decays gracefully but remains far above null at all tested horizons.

component count?

One-step transition taxonomy. At the one-step level, each embedded isolate can affect local component count via four transition types: (i) *focal death*: the isolate cell dies, removing a local connectivity node; (ii) *survival-connected merger*: the cell survives and fuses with an adjacent component, absorbing an independent fine component and reducing the count by one; (iii) *birth-mediated bridge*: a newly born neighbour cell connects the isolate to another component; and (iv) *local gain/fragmentation*: births around the isolate create new independent components, partially offsetting the loss. Type (i) is partially analytic: for single-diagonal isolates the focal cell has exactly one live Moore-neighbour, so it dies under S_{23} regardless of surrounding context (see below). Types (ii)–(iv) depend on the surrounding 5×5 neighbourhood configuration and are measured empirically.

We decompose iso_count into one-step outcome categories: death, survival, survival-connected (merges with a larger component), or triggering births at orthogonal or diagonal

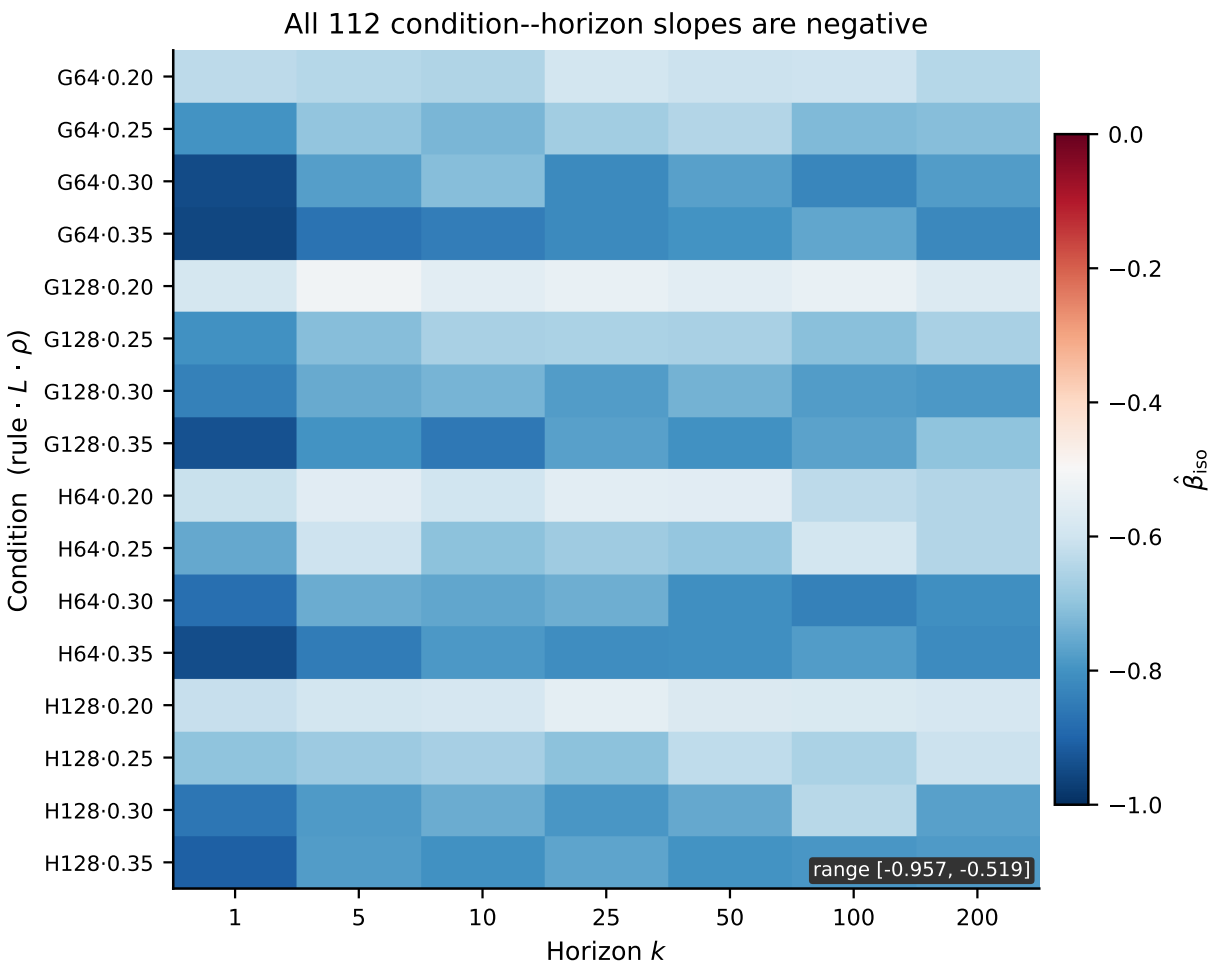

FIG. 4. **All 112 condition–horizon slopes are negative.** Full 16×7 heatmap of $\hat{\beta}_{\text{iso}}$. Rows are conditions (rule $\cdot$ L $\cdot$ ρ), columns are horizons $k \in \{1, 5, 10, 25, 50, 100, 200\}$. Every cell is negative, confirming the negative temporal response without exception.

sites. We also compute a *local-window* feature: the total component-count decrease within a 5×5 window centred on each isolate over one step. Cross-validated R^2 values for these features are shown in Fig. 5a.

The local-window feature dominates: $R^2_{\text{local_win}} = 0.538$, compared with $R^2_{\text{iso_count}} = 0.355$. The full fate set reaches $R^2_{\text{all_fates}} = 0.545$. Critically, the local-window gain (iso_local_window_gain_sum) has a positive slope, partially offsetting the loss: the net mechanistic pathway is *local component-context loss*, not simple cell death. Figure 5b shows the standardised slopes for each fate feature, revealing the partial cancellation between loss and gain.

Class-level analysis (Fig. 6) reveals a consistent physical story.

a. Exact deaths among orthogonal-degree-zero cells. Any alive cell with orthogonal degree 0 has Moore-neighbourhood count equal to its diagonal-neighbour count. Under S_{23}, survival requires 2 or 3 live Moore-neighbours; therefore: truly isolated cells (Moore degree 0, no live neighbours at all) die; single-diagonal embedded isolates (classes 1, 2, 4, 8; Moore degree 1) die; and four-diagonal isolates (class 15; Moore degree 4) also die. All three cases are exact consequences of the transition table, independent of surrounding context; death fractions of 1.0 for these classes (Fig. 6) are analytically guaranteed. Two- and three-diagonal isolates (Moore degrees 2 and 3) survive under S_{23} and depend on the surrounding 5×5 context for component merging. This analytic structure directly explains why the orthogonal-degree-zero superset outperforms the embedded-isolate subset in raw predictive R^2 (Table III): it combines both dying subclasses (Moore degree 0 and 1) under a single count. Embedded isolates remain the mechanistically preferred motif because the diagonal-tethering condition gives them a concrete component-context interpretation absent from truly isolated floating cells.

Two- and three-diagonal embedded isolates more often survive-connected: they merge with a larger component and reduce the fine component count by absorbing an independent component. Both pathways lead to the same aggregate outcome—local fine-connectivity loss—which is why the local-window loss feature ($R^2 = 0.538$) dominates over simple death counts ($R^2 = 0.355$ for iso_count alone). The dominant process is the disappearance of fine-net connectivity in the local neighbourhood, aggregated over many isolate classes and many conditions.

TABLE II. Mechanism R^2 decomposition at horizon $k = 1$ (global pooled). CV: 5-fold cross-validation.

Feature	CV R^2	Direction
iso_count (prestate)	0.355	–
local_window_loss	0.538	–
all_fates (combined)	0.545	mixed

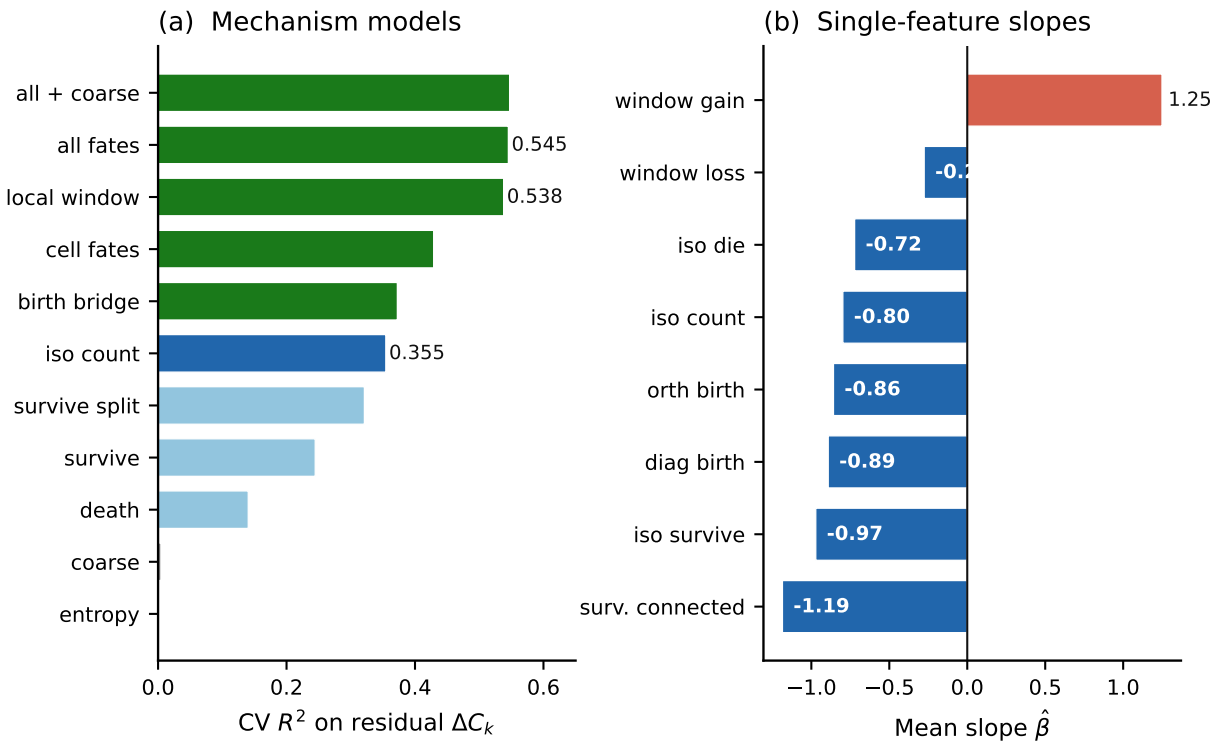

FIG. 5. **Mechanism carrier decomposition.** (a) Cross-validated R^2 for each fate and mechanism feature. The local-window loss feature ($R^2 = 0.538$) outperforms iso_count alone ($R^2 = 0.355$), identifying local component-context loss as the mechanistic carrier. (b) Standardised slopes: loss-related features are negative, gain-related features are positive, revealing partial cancellation.

VI. ROBUSTNESS: TRANSFER, AMPLITUDE, AND LOCAL BASELINES

The mechanism analysis tells us what the carrier is. Two robustness questions remain: does it generalise to held-out conditions, and does the scale of the response follow a systematic pattern?

A. Standardised Transfer

We z-score predictors and targets within each condition, then test whether the standardised mechanism coefficients transfer to held-out conditions under four leave-one-out axes (leave-density, leave-size, leave-rule, leave-condition). All four transfer axes pass (Fig. 7a). For the full fate representation, $R_z^2 = 0.544$ and frac R^2-positive $= 1.00$ on the leave-condition axis. iso_count alone also transfers ($R_z^2 = 0.355$). The standardised mechanism

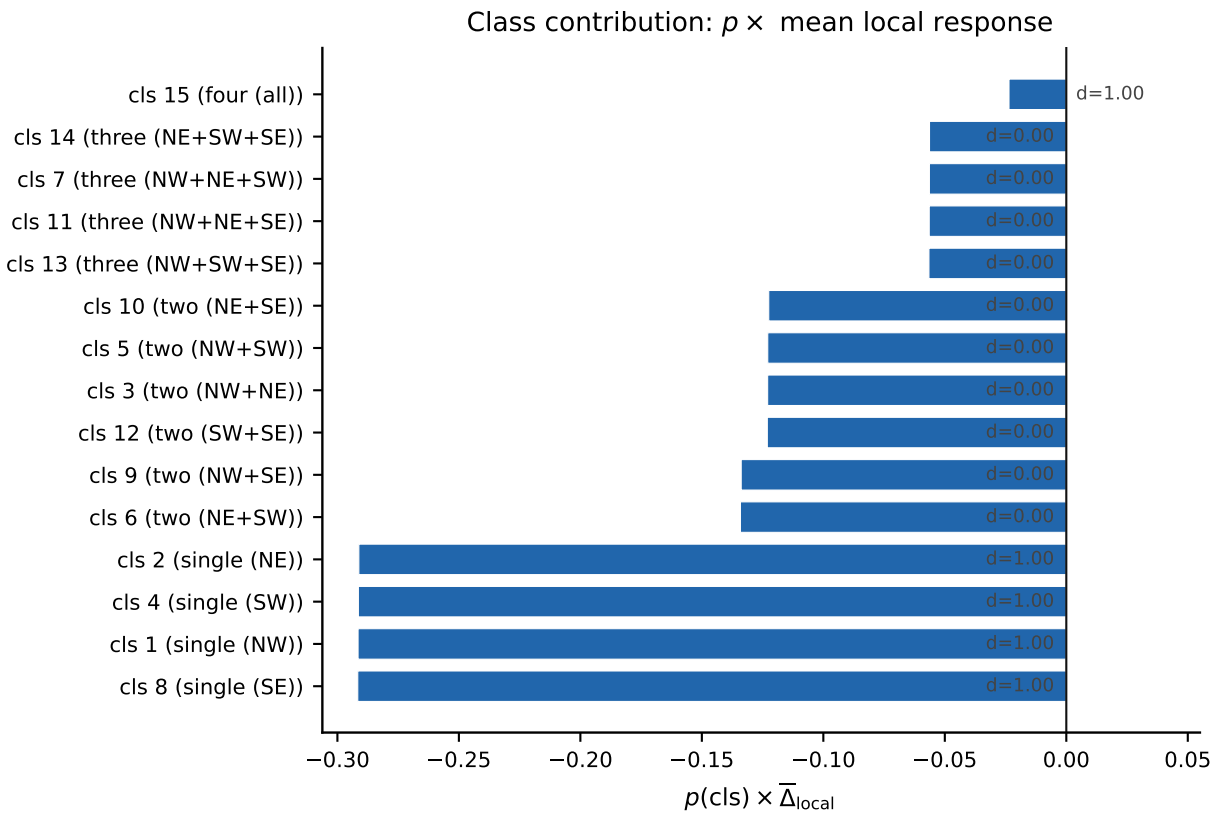

FIG. 6. **Class-level contributions to the mechanism.** Horizontal bars show $p(\text{cls}) \times \overline{\Delta}_{\text{local}}$ for each diagonal-adjacency class. Annotations give the death fraction (d) from GoL transition-class statistics. All classes contribute negatively; single-diagonal-neighbour classes drive the bulk via their high prevalence.

direction is condition-invariant: whichever leave-one-out split is applied, a model trained on the remaining conditions predicts the held-out condition well.

B. Amplitude Calibration

Condition amplitudes (raw residual slopes) vary substantially (amplitude CV = 0.369) but are predictable from grid size and density. The parsimonious size+ρ model—two predictors, centred $\log_2 L$ and centred ρ—achieves LOO $R^2 = 0.970$ on the 16 condition-level amplitudes. The full model (six predictors: centred $\log_2 L$, centred ρ, rule indicator, and three cross-terms) adds only marginal gain (LOO $R^2 = 0.977$). Size-only gives LOO $R^2 \approx 0.802$; rule-only gives -0.138, confirming that grid size L is the dominant predictor and rule iden-

tity alone is not. The two-predictor amplitude model uses 16 condition-level observations and two predictors; it should be read as a compact within-grid calibration rather than a universal scaling law.

The raw response thus decomposes as:

$$\hat{\beta}_{\text{iso}}(\text{condition}) \;\approx\; A(L,\rho)\,\hat{\beta}_{\text{iso},z}\,, \tag{1}$$

where $\hat{\beta}_{\text{iso},z} \approx -0.50$ is the transferable standardised slope and $A(L,\rho)$ is a condition amplitude calibrated from (L,ρ) with LOO $R^2 = 0.970$ for the two-predictor model. The natural explanation is that L sets the number of components and hence the scale of fine-component change, while ρ controls the density of the structural class; together they set the amplitude scale that multiplies the transferable standardised direction.

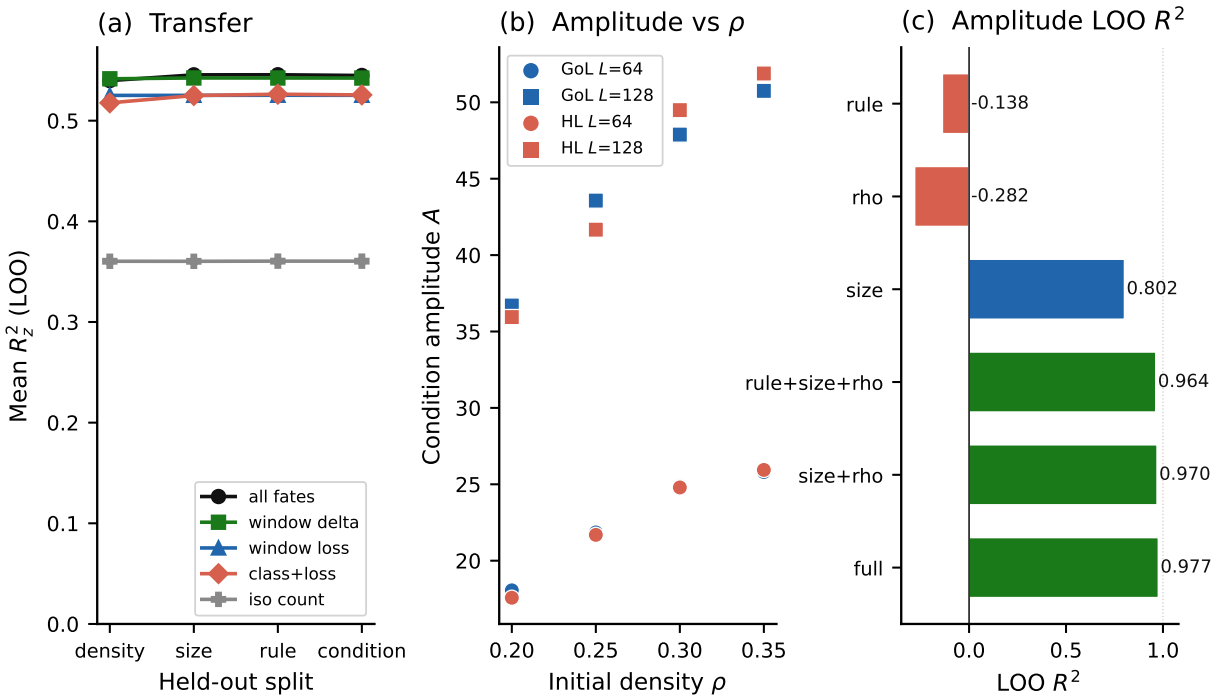

FIG. 7. **Transfer and amplitude calibration.** (a) Standardised transfer R_z^2 for five mechanism models across four leave-one-out axes; all lines stay positive. (b) Condition amplitude versus initial density ρ, coloured by grid size L and rule. Amplitude grows with L and is not determined by rule identity. (c) Amplitude LOO R^2 by model type; full model achieves $R^2 = 0.977$, size+ρ alone reaches 0.970.

C. Local Topology Baselines

Because the target is a change in 4-connected component count, embedded isolates might be merely one member of a broader family of local topological fragility statistics, each predicting ΔC_k for similar reasons. We test this directly by running the same density-residualised protocol (Appendix A) on fifteen alternative t=0 statistics: initial component count $C(0)$, small-component counts (sizes ≤ 4, $= 2, 3, 4$), singleton components (algebraically identical to orthogonal-degree-zero cells), truly isolated cells (Moore degree 0), Moore-degree counts ($d = 1, 2$), boundary cells (alive with ≥ 1 orthogonally dead neighbour), and coarse block statistics. Table III shows global mean residual R^2 per predictor, averaged over all 112 condition-horizon cells.

The result is structurally informative and transparent. $C(0)$ is the strongest predictor ($R^2 = 0.352$): more components at t=0 means more components "at risk" of loss. Small components of size ≤ 4 are next ($R^2 = 0.330$), followed by singleton components / orthogonal-degree-zero cells ($R^2 = 0.274$). Embedded isolates rank fourth ($R^2 = 0.234$), behind the orthogonal-degree-zero superset from which they are drawn. All block-level and Moore-degree-only statistics are substantially weaker.

Embedded isolates are therefore not claimed to be the unique or globally optimal predictor. They are a compact, diagonal-tethered subclass of the orthogonal-degree-zero family, singled out because their one-step fate has a clean analytic form under S_{23} and because the diagonal-tethering condition provides a concrete component-context interpretation: a cell touching live structure diagonally but not orthogonally is a fragile link whose death or merger directly removes a fine-connectivity node. The broader local-fragility family—from $C(0)$ to small-component counts to singleton cells to embedded isolates—forms a coherent predictive structure that this paper characterises and places on a mechanistic footing.

VII. DISCUSSION

Summary. Local topological fragility—the initial density of weakly-connected live cells—predicts future fine-component loss in GoL and HighLife. The topology-baseline battery (Table III) places this on a tested footing: the predictive hierarchy from $C(0)$ through small-component counts and singleton/orthogonal-degree-zero cells to embedded isolates

TABLE III. Local topology baseline: mean residual R^2 per predictor, averaged over all 112 condition-horizon cells (2 rules $\times$ 2 sizes $\times$ 4 densities $\times$ 7 horizons; 1000 worlds per condition). All predictors measured at t=0; live-count + density baseline residualised before fitting. Singleton components $\equiv$ orthogonal-degree-zero cells by definition.

Predictor	Mean resid. R^2
$C(0)$ (initial component count)	0.352
Small components ($\leq$ 4 cells)	0.330
Singleton / orth-degree-zero cells	0.274
Embedded isolates (iso_count)	**0.234**
Moore-degree-one cells	0.033
Block variance / entropy	$\approx$0.006–0.009

($R^2 = 0.234$) forms a coherent fragility family, with embedded isolates as the mechanistically transparent motif. The target-specific prestate signal ($R^2 = 0.241$; both shuffle nulls near zero) arises via a partial analytic mechanistic pathway: any orthogonal-degree-zero cell with Moore degree 0 or 1 dies exactly under S_{23} (analytic); multi-diagonal isolates more often survive-connected and generate local-window loss (empirical). The standardised response transfers across all leave-one-out axes, with raw amplitude calibrated from (L, ρ) (LOO $R^2 = 0.970$).

Common-direction diagnostic. A descriptive check on why a single statistic serves all fine-net horizons is to measure alignment of regression-coefficient vectors in feature space. For each task we fit a residual model and normalise the coefficient vector; pairwise absolute cosines $|\cos\theta_{ij}|$ measure task alignment. Fine-net horizon tasks are near-collinear ($|\cos| = 0.999$, rank-1 energy = 0.999): a single feature direction aligns all k-step predictions simultaneously. Heterogeneous targets are not ($|\cos| = 0.427$), and no single predictor adds meaningful ΔR^2 for all of them. Figure 8 shows this contrast. This collinearity is a descriptive geometric property of the regression directions, not a claim about any specific theory of causal emergence or effective coarse-graining.

The two-layer structure. The decomposition $\hat{\beta}_{\text{iso}}(\text{cond}) \approx A(L, \rho)\, \hat{\beta}_{\text{iso}}, z$ cleanly separates the transferable standardised mechanism from the condition-specific scale. This matters for applications: if a practitioner has access to a new (L, ρ) condition, they can predict the raw

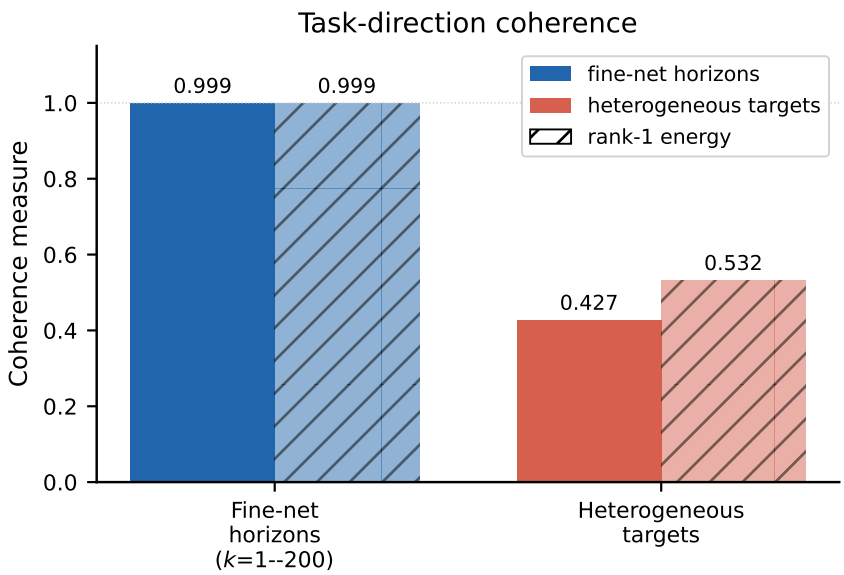

FIG. 8. **Task-direction coherence.** Mean pairwise $|\cos\theta|$ (solid) and rank-1 energy (hatched) for fine-net horizon tasks (blue) and heterogeneous targets (orange). Fine-net horizons are near-collinear ($|\cos| = 0.999$, rank-1 $= 0.999$); heterogeneous targets are dispersed ($|\cos| = 0.427$).

slope amplitude with LOO $R^2 \approx 0.970$ from (L, ρ) alone, and they can apply the standardised slope direction directly from the training pool. The two layers are independently portable.

What is not earned. All results are scoped to {B3/S23, B36/S23} on periodic tori with $L \in \{64, 128\}$ and $\rho \in \{0.20, 0.25, 0.30, 0.35\}$. Extension to other Life-like rules, larger grids, continuous-state CAs, or non-random initial conditions is an open empirical question. Closed-form derivation of $\hat{\beta}_{\text{iso}}$ from microscopic transition rules is not provided. Raw amplitude transfer without per-condition calibration is not supported: amplitude varies with (L, ρ) by a factor of ≈ 0.369 (CV).

Rule dependence of the embedded-isolate motif. The exact one-step deaths established here—Moore-degree-0 and Moore-degree-1 orthogonal-degree-zero cells dying under S_{23}—rely on the specific survival set $S = \{2, 3\}$ shared by GoL and HighLife. A Life-like rule with a different survival set would alter or eliminate these exact fates and change the composition of the local-fragility hierarchy: the embedded-isolate motif is rule-dependent through its dependence on S_{23}. The broader local-fragility concept—that the initial density of weakly-connected live cells predicts future fine-component loss—may generalise to other Life-like rules, but whether $C(0)$, small-component counts, or embedded isolates retain their relative

rankings under alternative survival sets is an open empirical question and a natural extension of this work.

Relation to structural and computational work. Component-size and local-connectivity statistics have appeared as descriptors in CA pattern analyses[5], but their systematic comparison under a common density-residualised predictive protocol—revealing a coherent local-fragility hierarchy from $C(0)$ through small-component counts to embedded isolates—is, to our knowledge, new. The hierarchy parallels intuitions from percolation theory, where component-size distributions characterise connectivity near occupation thresholds, but the focus here is dynamical: which t=0 local counts predict future change in fine-scale component number. Local causal-state methods[4,5] infer predictive-equivalence partitions from joint past/future lightcone statistics; embedded isolates are a hand-defined structural class within that broader space: they are not claimed to be a full predictive-equivalence partition, but they capture one target-specific component of future fine-connectivity change with a simple geometric definition and without requiring lightcone data or a learned partition. The amplitude calibration does not conflict with coarse-graining impossibility results[6]: we predict one low-dimensional target statistic (ΔC_k), not the full future CA configuration. The partial predictability shown here is consistent with computational irreducibility; it concerns a specific chosen observable, not global state reconstruction. The novelty lies in combining four elements simultaneously: a coherent local-fragility family that (i) is characterised by a baseline battery rather than a single hand-picked object, (ii) has a mechanistically transparent motif (embedded isolates) with exact one-step fate for the single-diagonal subclass, (iii) produces a standardised mechanism that transfers across held-out conditions, and (iv) admits amplitude calibration from (L, ρ)—all for future fine-component loss in two well-studied Life-like rules.

Local topological fragility as an observer class. The topology-baseline battery shows that what matters for predicting fine-component change is the initial density of locally disconnected or weakly-connected live cells—a property we call local topological fragility. $C(0)$, small-component counts, and orthogonal-degree-zero cells all capture this property in different ways and form a coherent predictive family. Embedded isolates are the analytically cleanest member of that family: their diagonal tethering gives them a component-context interpretation (they are nodes at risk of losing their local connectivity), and their one-step fate is partially analytic rather than purely empirical. The stability of $\hat{\beta}_{\text{iso}}(k)$ across hori-

zons suggests that the risk resolves primarily at $t = 1$ and that subsequent dynamics do not erase the initial predictive information—consistent with the prestate non-leakiness result. iso_count is therefore a *topological risk indicator* within the broader local-fragility family, not a simple density proxy and not claimed to be the uniquely optimal fragility statistic.

VIII. CONCLUSION

Future fine-component change in Conway's Game of Life and HighLife is controlled by local topological fragility: the initial density of weakly-connected live cells. A topology-baseline battery of sixteen statistics identifies a predictive hierarchy from the initial component count $C(0)$ ($R^2 = 0.352$) through small-component and singleton/orth-degree-zero counts ($R^2 = 0.330$, 0.274) to embedded isolates ($R^2 = 0.234$), the most analytically transparent motif. The $t = 0$ embedded-isolate count alone explains a substantial fraction of residual fine-component change across all tested horizons (mean prestate $R^2 = 0.241$; minimum $R^2 = 0.175$ at $k = 200$; shuffle ≈ 0.0008). The residual slope $\hat{\beta}_{\text{iso}}(k)$ is negative in all 112 condition-horizon combinations (7 horizons $\times$ 16 conditions; 2 rules $\times$ 2 sizes $\times$ 4 density levels), an empirical regularity that holds without exception across the scoped battery. Both iso-shuffle and target-shuffle incremental nulls are near zero ($\Delta R^2 < 0.001$), ruling out density artefacts. The one-step mechanistic pathway is local component-context loss, not simple cell death: any orthogonal-degree-zero cell with Moore degree 0 or 1 dies exactly under the shared S_{23} rule; two- and three-diagonal isolates often survive-connected. This standardised mechanism transfers across all leave-one-out axes and the raw amplitude is predictable from (L, ρ) (LOO $R^2 = 0.970$, rising to 0.977 in the full six-predictor model).

The collinearity of fine-net regression directions ($|\cos| = 0.999$) provides a compact geometric diagnostic of why one prestate statistic serves all tested horizons simultaneously.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

All code, simulation outputs, and figure sources are available at `https://github.com/kunalb541/CA`. The repository includes an automated regression suite (93 tests) covering the selection, horizon-response, mechanism, transfer, amplitude-calibration, and artifact-generation pipelines, plus a standalone topology-baseline controls script (`ca_topology_baseline_controls.py`) whose outputs are archived in `outputs/topology_baselines/`; all regression tests pass for the submitted build.

Appendix A: Regression Details and Null Definitions

Residualisation protocol

For each prediction world, the density nuisance covariate is $N_{\text{live}}(0)$, the total live-cell count at $t = 0$ (proportional to ρ up to Bernoulli sampling variance). All regression models include $N_{\text{live}}(0)$ as a linear covariate before adding iso_count or any fate feature. $\Delta R^2 = R^2(\text{density + predictor}) - R^2(\text{density only})$. The residual slope $\hat{\beta}_{\text{iso}}$ is the OLS coefficient on iso_count in this density-residualised fit.

Null definitions

Two nulls are used: (i) *iso-shuffle null*: iso_count labels are permuted uniformly at random within each condition; ΔR^2 and slope are recomputed. (ii) *target-shuffle null*: the target ΔC_k is permuted; ΔR^2 is recomputed with the original iso_count. Both converge near zero ($\Delta R^2 < 0.001$), confirming the signal depends on the joint alignment of iso_count with ΔC_k and is not a density or permutation artefact.

R^2 notation and computation contexts

Six distinct R^2 quantities appear in this paper; they answer different questions and should not be compared directly. Table **??** provides a concise reference.

TABLE IV. R^2 quantities used in this paper, their computation, and the sections in which they appear. All fits are OLS unless noted. "Condition-wise" means fitted independently within each of the 16 rule×size×density batteries.

Quantity	Computation	Sections
Ordinary R^2	OLS variance explained; no cross-validation; used for prestate fits	III
Residual R^2	R^2 of iso_count after $N_{\text{live}}(0)$ projected out; condition-wise	III, IV, VI C
ΔR^2	R^2(density+predictor) − R^2(density only); condition-wise target-selection screen	III
CV R^2	5-fold cross-validated R^2; condition-wise; used for mechanism features	V
R^2_z (standardised)	OLS on z-scored predictors and targets across held-out conditions (LOO axes)	VI A
LOO R^2	Leave-one-out cross-validation on the 16 condition-level amplitudes; two predictors ($\log_2 L$, ρ)	VI B

Appendix B: Transition-Class Coding

Diagonal-adjacency class is an integer in $[1, 15]$ indexing the bit pattern of the four diagonal neighbours of an embedded isolate (NW, NE, SW, SE), with class c equal to the binary value $NW \cdot 1 + NE \cdot 2 + SW \cdot 4 + SE \cdot 8$. Class 15 (all four diagonal neighbours alive) is rare in this battery ($p \approx 0.009$, $n = 80{,}015$ instances) but contributes a negative local response ($\bar{\Delta} \approx -2.52$, death fraction = 1.0) comparable to other multi-diagonal classes; it is not excluded from the analysis. Classes 1, 2, 4, 8 are the *single-diagonal* classes; classes 3, 5, 6, 9, 10, 12 are the *two-diagonal* classes; classes 7, 11, 13, 14 are the *three-diagonal*

classes; class 15 is the *four-diagonal* class.

Appendix C: Background Observer-Scale Diagnostics

The main text establishes the embedded-isolate temporal response using the 16-condition GoL/HighLife battery described in Section II. For lineage and context, this appendix records earlier observer-scale diagnostics from the same research program. These diagnostics show *why* a target-indexed description analysis was a natural next step: fine and coarse observers disagree systematically, and different targets are optimally predicted at different descriptive scales. **These figures are background diagnostics only; they are not load-bearing evidence for the response estimates in the main text.**

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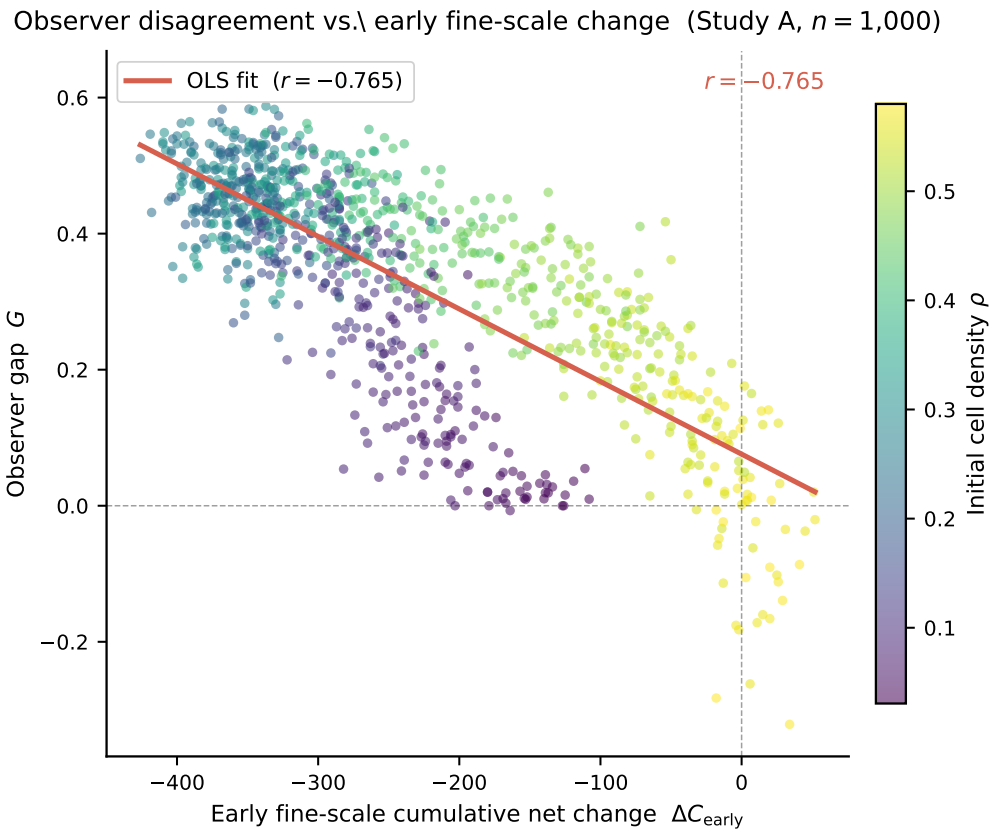

FIG. 9. **E1 — Observer disagreement scatter (Study A, $n = 1{,}000$).** Each point is one simulated GoL world. x-axis: early fine-scale cumulative net component change ΔC_{early}; y-axis: observer gap G (fine minus coarse observer agreement); colour: initial cell density ρ. The regression line gives $r = -0.765$ ($p \ll 10^{-100}$); partial r controlling for density $= -0.848$. Worlds where fine-scale structure degrades early tend to produce larger observer disagreement. *Background diagnostic only; not load-bearing for the response estimates in the main text.*

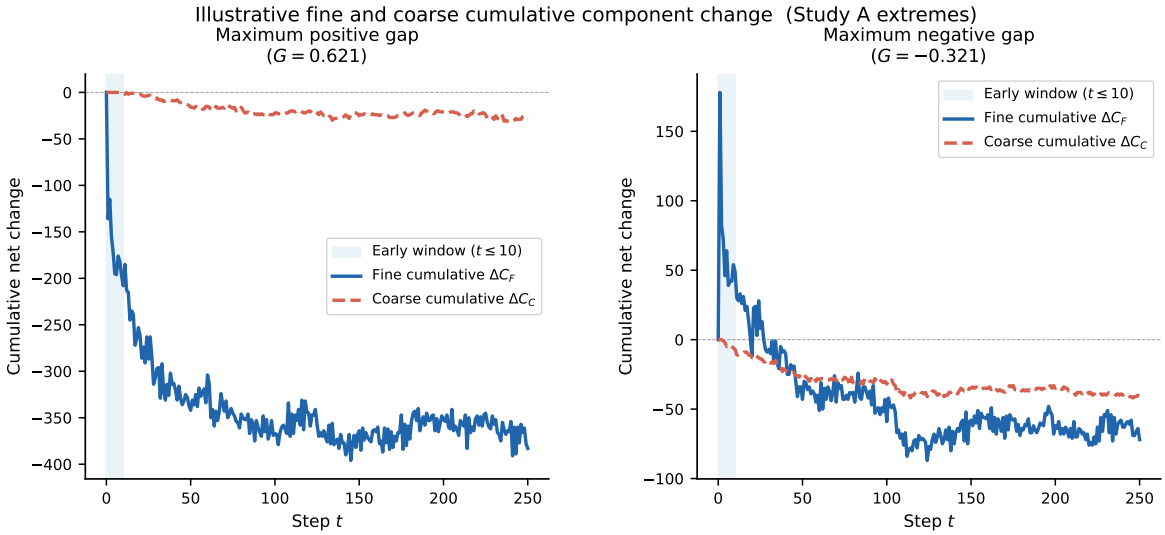

FIG. 10. **E2 — Illustrative fine and coarse observer trajectories (Study A extremes).** Left: the world with the maximum positive observer gap ($G = +0.621$). Right: the world with the maximum negative gap ($G = -0.321$). Blue solid: fine-scale cumulative net component change $\Delta C_F(t)$. Orange dashed: coarse-scale cumulative net change $\Delta C_C(t)$. Shaded region: early window ($t \leq 10$) whose summary statistic ΔC_{early} is used in Figure 9. These are selected extreme cases, not representative samples. *Background diagnostic only; not load-bearing for the response estimates in the main text.*

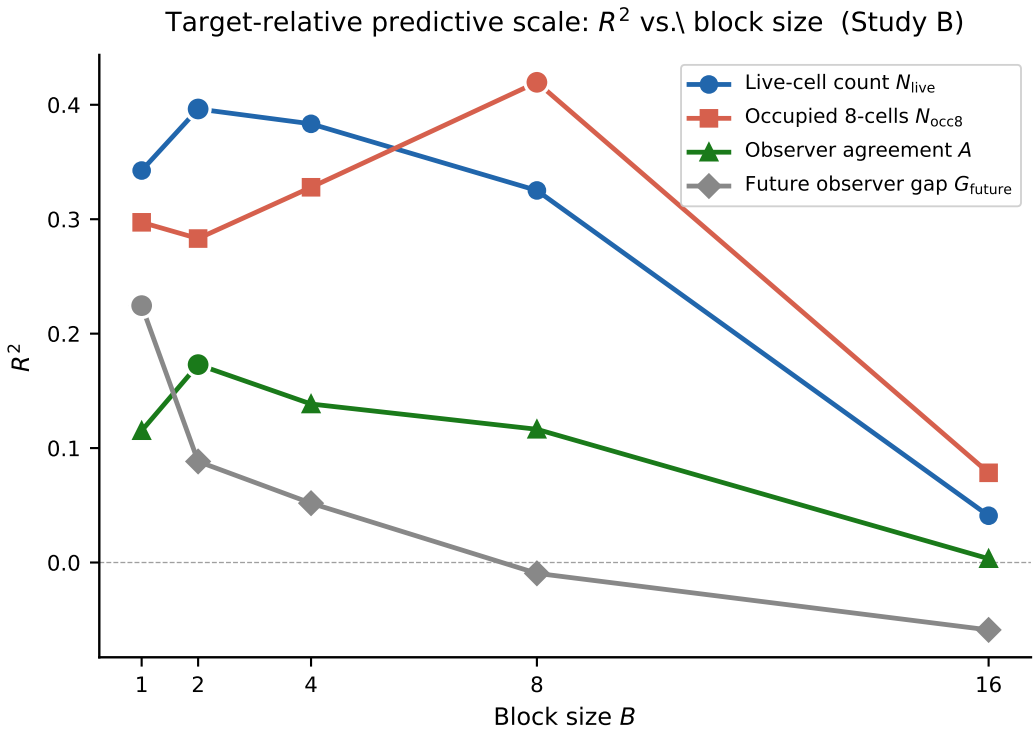

FIG. 11. **E3 — Target-relative predictive scale (Study B).** R^2 of a block-aggregated feature predictor as a function of block size B, separately for four prediction targets. N_{live} peaks at $B = 2$; N_{occ8} peaks at $B = 8$; future observer gap G_{future} peaks at $B = 1$ and falls steeply; observer agreement A is broadly flat then drops. Different targets favour different descriptive scales, motivating the target-specific framing of the main-text analysis. *Background diagnostic only; not load-bearing for the response estimates in the main text.*

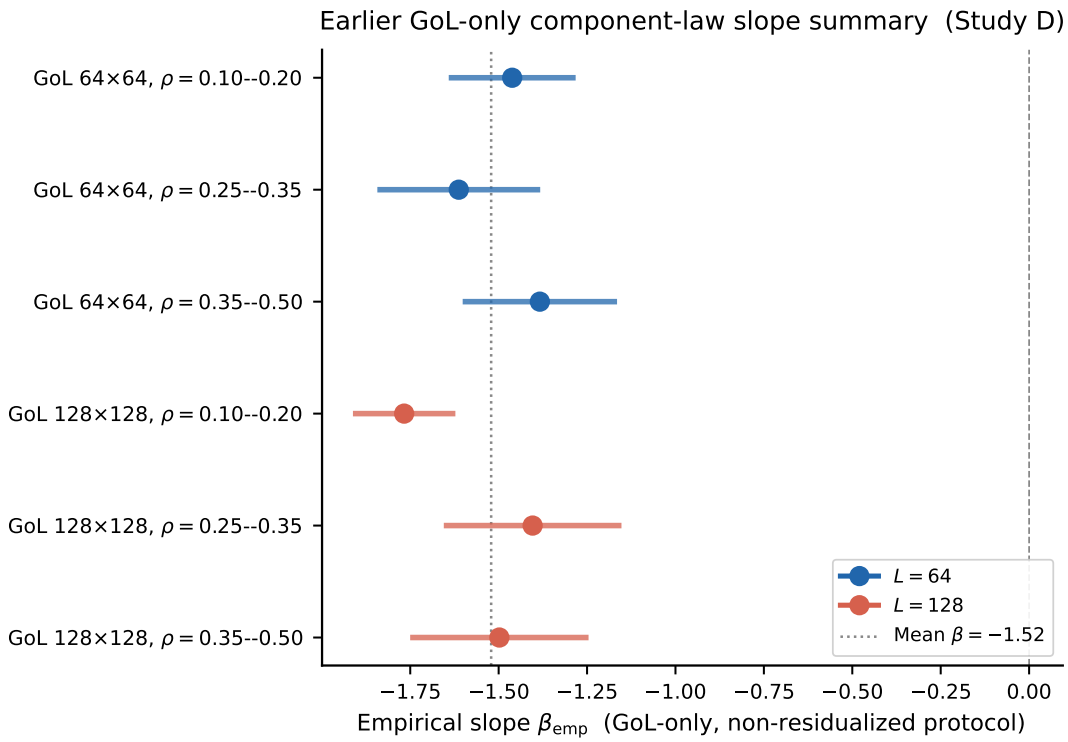

FIG. 12. **E4 — Earlier GoL-only component-response slope summary (Study D).** Empirical slopes β_{emp} with 95% CI for six GoL conditions (two grid sizes, three density bands) under an earlier, non-residualized regression protocol. Mean $\beta \approx -1.52$. **This coefficient is not directly comparable to the residualized slopes $\beta_{\text{iso}} \approx -0.70$ in the main text:** the two studies use different targets, different residualization, and different simulation batteries. This figure is included as historical lineage only. *Background diagnostic only; not load-bearing for the response estimates in the main text.*